Assignment 8

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Please note that

- **Released date: November 29th, Friday.**
- **Due date: December 12th, Thursday, by 11:55pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS6.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

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Question 1. Let u be a unit vector in $\mathbb{R}^n$ and let $H = I - 2uu^T$. Show that H is both orthogonal and symmetric and hence is its inverse.

Question 2. Give $x_1 = \frac{1}{2}(1,1,1,-1)^T$ and $x_2 = \frac{1}{6}(1,1,3,5)^T$, verify that these vectors form an orthonormal set in $\mathbb{R}^4$. Extend this set to an orthonormal basis for$\mathbb{R}^4$ by finding an orthonormal basis for the null space of

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$$\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{bmatrix}$$

Hint: First find a basis for the null space and then use the Gram-Schmidt process.

Question 3. Find the eigenvalues and the corresponding eigenspaces for each of the following matrices:

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(a) $\begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 5 & -1 \end{bmatrix}$

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Question 4. Let A be an $n \times n$ matrix. Prove that A is singular if and only if $\lambda = 0$ is an eigenvalue of A.

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Question 5. Let A be a nonsingular matrix and let λ be an eigenvalue of A. Show that $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}.

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Question 6. An $n \times n$ matrix A is said to be idempotent if $A^2 = A$. Show that if λ is an eigenvalue of an independent matrix, then λ must be either 0 or 1.

Question 7. Let A be an $n \times n$ matrix and let $B = A + I$.Is it possible for A and B to be similar? Explain.

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Question 8. Let Q be an orthogonal matrix. 8_13

(a) Show that if λ is an real eigenvalue of Q, then $|\lambda| = 1$.

(b) Show that $|\det(Q)| = 1$.

Question 9. Show that if two $n \times n$ matrix A and B have a common eigenvector x (but not necessary a common eigenvalue), then x will also be an eigenvector of any matrix of the form $C = \alpha A + \beta B$.

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Question 10. In each of the following,factor the matrix A into a product XDX^{-1}, where D is diagonal:

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(a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$

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Question 11. For each of the matrices in the above Exercise 10, use the XDX^{-1} factorization to compute A^6

Question 12. An $n \times n$ matrix is said to be nipotent if $A^k = O$ for some positive integer k. Show that all eigenvalues of a nilpotent matrix are 0.

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Question 13. Find the characteristic polynomials, the eigenvalues and the eigenspaces of the following matrices:

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(a) $\begin{bmatrix} 7 & -4 \\ -8 & -7 \end{bmatrix}$

(b) $\begin{bmatrix} 5 & 2 & 3 \\ -13 & -6 & -11 \\ 4 & 2 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 1 & 1 \\ -15 & -5 & -5 \\ 6 & 2 & 2 \end{bmatrix}$

Question 14. Which of the matrices

$$\begin{pmatrix} 11 & 5 & 8 \\ -30 & -16 & -30 \\ 9 & 5 & 10 \end{pmatrix}, \begin{pmatrix} -2 & -1 & -1 \\ 9 & 4 & 3 \\ -3 & -1 & 0 \end{pmatrix}, \begin{pmatrix} -4 & -1 & 1 \\ 17 & 4 & -5 \\ -5 & -1 & 2 \end{pmatrix}$$

are similar to diagonal matrices?

Question 15. Let $A \in \mathbb{R}^{n\times n}$,$A^2 - 3A + 2I_n = 0$, assume λ is eigenvalue of A, show that $\lambda = 1$ or 2.